

# Supplementary Material

## Learned initialization for variational quantum circuits on lattice and field-theory Hamiltonians: a ten-system benchmark of Q-MAML

Arnav Kapoor

This document collects the per-seed numbers behind the means reported in the main text, and three exploratory results that motivated choices made there but are not needed to follow its argument.

### S1 Per-seed results for the ten Hamiltonians

Table S1 gives the raw per-seed relative error (or log-gap, Track F) underlying Table 2 of the main text: every number that mean and win/tie/loss counts there are computed from.

Table S1: Per-seed relative error (%; log-gap for Track F) for every scheme and system at base size.

| Track | System                                | Seed | Q-MAML               | Zero                 | $\pi$                | Uniform              | Gaussian             |
|-------|---------------------------------------|------|----------------------|----------------------|----------------------|----------------------|----------------------|
| A     | Schwinger, $L=2$                      | 0    | $2.0 \times 10^{-3}$ | 396                  | 0.056                | 1.61                 | 0.194                |
|       |                                       | 1    | 0.277                | 245                  | 0.288                | 2.14                 | 0.529                |
|       |                                       | 2    | 0.264                | 292                  | 0.304                | 1.40                 | 0.636                |
| B     | $\mathbb{Z}_2$ LGT, penalty ansatz    | 0    | 59.16                | 112                  | 24.69                | 23.07                | 24.05                |
|       |                                       | 1    | 30.01                | 110                  | 31.88                | 15.18                | 21.37                |
|       |                                       | 2    | 58.96                | 113                  | 25.47                | 22.76                | 12.88                |
| B     | $\mathbb{Z}_2$ LGT, gauge-inv. ansatz | 0    | $3.1 \times 10^{-3}$ | 0.016                | 0.027                | 0.022                | 0.015                |
|       |                                       | 1    | $2.9 \times 10^{-3}$ | $9.6 \times 10^{-3}$ | $8.8 \times 10^{-3}$ | $9.0 \times 10^{-3}$ | 0.010                |
|       |                                       | 2    | $6.6 \times 10^{-4}$ | $8.5 \times 10^{-3}$ | $7.5 \times 10^{-3}$ | $7.3 \times 10^{-3}$ | 0.012                |
| C     | $\phi^4$ scalar field                 | 0    | 5.86                 | 5.77                 | 36.06                | 7.11                 | 7.55                 |
|       |                                       | 1    | 4.19                 | 4.42                 | 14.58                | 4.60                 | 4.74                 |
|       |                                       | 2    | 8.11                 | 8.16                 | 30.38                | 9.00                 | 8.26                 |
| D     | SU(2) LGT, $N=2$                      | 0    | $2.1 \times 10^{-5}$ | 1390                 | 0.020                | 0.038                | 0.023                |
|       |                                       | 1    | $9.5 \times 10^{-6}$ | 775                  | 0.011                | $8.5 \times 10^{-3}$ | 0.012                |
|       |                                       | 2    | $9.7 \times 10^{-6}$ | 1767                 | 0.015                | 0.019                | $5.9 \times 10^{-3}$ |
| E     | Neutrinos, $N=4$                      | 0    | $2.2 \times 10^{-5}$ | 0.016                | $4.1 \times 10^{-4}$ | $4.4 \times 10^{-3}$ | $2.7 \times 10^{-3}$ |
|       |                                       | 1    | 0.011                | 0.078                | $5.1 \times 10^{-3}$ | 0.096                | 0.051                |
|       |                                       | 2    | $8.0 \times 10^{-5}$ | 0.073                | $7.2 \times 10^{-4}$ | $7.8 \times 10^{-4}$ | $4.1 \times 10^{-3}$ |
| F     | SUSY QM (log-gap)                     | 0    | -4.87                | 0.00                 | -2.40                | -4.75                | -4.66                |
|       |                                       | 1    | -4.90                | -0.58                | -3.28                | -4.62                | -4.62                |
|       |                                       | 2    | -3.67                | -0.05                | -1.95                | -4.89                | -4.35                |
| G     | $\mathbb{D}_8$ LGT                    | 0    | 10.49                | 95.27                | 26.01                | 11.86                | 10.97                |

| Track | System               | Seed | Q-MAML               | Zero                 | $\pi$                | Uniform              | Gaussian             |
|-------|----------------------|------|----------------------|----------------------|----------------------|----------------------|----------------------|
| H     | Light nuclei (UCCSD) | 1    | 15.47                | 92.37                | 21.99                | 13.94                | 19.60                |
|       |                      | 2    | 6.47                 | 94.04                | 16.31                | 6.35                 | 5.67                 |
|       |                      | 0    | 0.885                | 0.885                | 0.885                | 0.885                | 0.885                |
|       |                      | 1    | 1.41                 | 1.41                 | 1.41                 | 1.41                 | 1.41                 |
| I     | Scalar Yukawa        | 2    | 2.02                 | 2.02                 | 2.02                 | 2.02                 | 2.02                 |
|       |                      | 0    | $2.7 \times 10^{-5}$ | $3.0 \times 10^{-5}$ | 1.37                 | $3.0 \times 10^{-4}$ | $3.7 \times 10^{-3}$ |
|       |                      | 1    | $2.7 \times 10^{-5}$ | $2.9 \times 10^{-5}$ | 0.550                | $3.3 \times 10^{-4}$ | $2.7 \times 10^{-3}$ |
|       |                      | 2    | $2.6 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | 2.42                 | $5.6 \times 10^{-5}$ | $3.4 \times 10^{-4}$ |
| J     | LMG, $J=3/2$         | 0    | $1.9 \times 10^{-6}$ | $1.1 \times 10^{-6}$ | $1.3 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | $2.2 \times 10^{-6}$ |
|       |                      | 1    | $5.5 \times 10^{-6}$ | $1.1 \times 10^{-6}$ | $2.3 \times 10^{-6}$ | $4.5 \times 10^{-6}$ | $2.9 \times 10^{-6}$ |
|       |                      | 2    | $6.2 \times 10^{-7}$ | $2.9 \times 10^{-6}$ | $3.3 \times 10^{-6}$ | $1.3 \times 10^{-6}$ | $6.5 \times 10^{-6}$ |
|       |                      |      |                      |                      |                      |                      |                      |

## S2 $\mathbb{Z}_2$ gauge theory: penalty-strength sweep

The main text’s section on structural gauge invariance states that, on the penalty-enforced hardware-efficient ansatz, Q-MAML is behind the best classical scheme at every penalty strength  $V$  tested, refuting the explanation that a large penalty simply drowns the task signal. Table S2 gives the single-seed final energies behind that statement (reduced 4-task, 200-iteration budget,  $V \in \{1, 3, 10, 30\}$ ); Q-MAML ties Uniform only at the weakest penalty and falls further behind as  $V$  grows to 10, recovering slightly by  $V = 30$ .

Table S2:  $\mathbb{Z}_2$  gauge theory, penalty-enforced ansatz: final energy by penalty strength  $V$  (single seed, reduced budget).

| $V$ | Q-MAML | Zero  | $\pi$ | Uniform | Gaussian | Q-MAML / best classical |
|-----|--------|-------|-------|---------|----------|-------------------------|
| 1   | 1.468  | 2.490 | 1.560 | 1.438   | 1.516    | 1.02 (Uniform)          |
| 3   | 2.683  | 3.335 | 2.280 | 1.655   | 2.042    | 1.62 (Uniform)          |
| 10  | 3.791  | 4.432 | 2.917 | 2.850   | 2.891    | 1.33 (Uniform)          |
| 30  | 4.157  | 5.498 | 3.729 | 3.657   | 4.165    | 1.14 (Uniform)          |

## S3 Lattice Schwinger model: the $L=4$ depth/pretraining-rate scan

The main text’s section on system size reports that depth, not pretraining convergence, decides the  $L=4$  (8-qubit) result. Table S3 gives the single-seed grid this was read off: pretraining learning rate 0.005 stalls (final energy far from the ground state at every depth-12 setting, visible as the shared Zero/Uniform/Gaussian columns not moving with epoch count), and at depth 12 no learning-rate or epoch-count change brings Q-MAML ahead of  $\pi$ ; only the depth-18 ansatz does, and by a wide margin.

Table S3: Lattice Schwinger model,  $L=4$ : final relative error (%) by depth and pretraining setting (single seed, reduced budget).

| Pretraining setting        | Q-MAML | Zero | $\pi$ | Uniform | Gaussian |
|----------------------------|--------|------|-------|---------|----------|
| Depth 12, 40 ep, lr 0.005  | 24.49  | 451  | 17.95 | 39.44   | 38.40    |
| Depth 12, 100 ep, lr 0.005 | 23.89  | 451  | 17.95 | 39.44   | 38.40    |
| Depth 12, 100 ep, lr 0.002 | 21.21  | 451  | 17.95 | 39.44   | 38.40    |
| Depth 18, 100 ep, lr 0.002 | 2.43   | 451  | 8.65  | 10.95   | 11.84    |

## S4 Quark–gluon jet tagging

A second classification dataset, quark–gluon jet tagging (six qubits, two seeds), is mentioned in the main text’s HEP classification section as directional evidence only. Table S4 gives the full comparison: Q-MAML’s validation accuracy and meta-loss against the three classical baselines used elsewhere in classification (the first-order variant was not run on this dataset).

Table S4: Quark–gluon jet tagging, six qubits ( $n = 2$  seeds): final validation accuracy and meta-loss.

| Scheme   | Validation accuracy | Meta-loss         |
|----------|---------------------|-------------------|
| Zero     | $0.594 \pm 0.021$   | $0.693 \pm 0.004$ |
| Uniform  | $0.594 \pm 0.021$   | $0.701 \pm 0.007$ |
| Gaussian | $0.599 \pm 0.036$   | $0.689 \pm 0.000$ |
| Q-MAML   | $0.776 \pm 0.005$   | $0.308 \pm 0.003$ |

## S5 Software and compute

Circuits are simulated with PennyLane; the Learner, encoders and optimizers use PyTorch (Adam throughout, double precision for the Learner). All results are noiseless statevector simulation; no quantum hardware was used. Exact ground energies and the symmetry checks in Table 1 (main text) use NumPy/SciPy dense or sparse diagonalization, run once per Hamiltonian module before any Q-MAML run (`verify_construction()` in each `hamiltonians_*.py`). Code, notebooks and saved results are at the repository given in the main text’s Data and code availability statement.